

1 Dimensionality Reduction with PCA

MNIST consists of 70,000 images in a 784-dimensional space. PCA was used to compress the data before clustering. The cumulative variance rises sharply within the first 50–80 components, after which returns diminish.

Variance Retained	Components
90%	~ 87
95%	~ 154
98%	~ 260
99%	~ 331

Table: PCA variance thresholds.

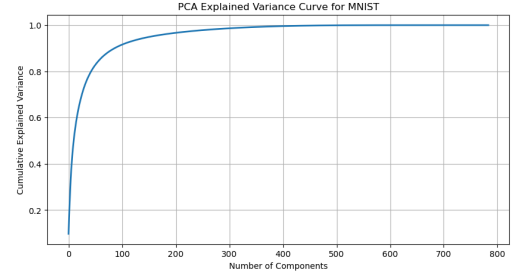


Figure 1: Cumulative explained variance curve.

The subsequent experiments investigate how this dimensionality reduction affects clustering performance across different models and metrics.

2 K-Means Clustering

2.1 High-Dimensional PCA Experiments (100–250)

K-Means was first evaluated with PCA dimensions between 100 and 250 using:

Parameter	Values Tested
k	8, 10, 12, 14
Initialisation	k-means++ , random
n_{init}	10, 20
Algorithm	lloyd , elkan

K-Means hyperparameter grid

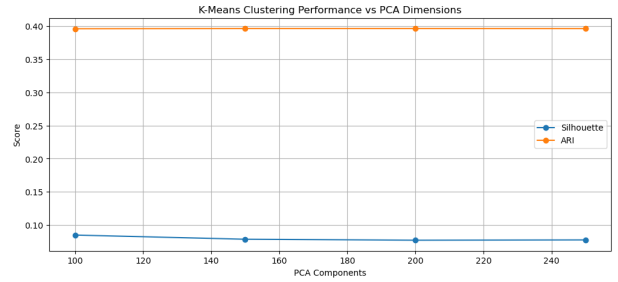


Figure 2: K-Means performance across high PCA dimensions

Across all PCA dimensions, the best models consistently used $k = 8$, and **k-means++** gave more stable solutions than random initialisation. Silhouette scores remained low and decreased slightly as PCA dimensionality increased. Although high-dimensional PCA captures more variance, it also reintroduces noise-like variation that weakens geometric cluster separation. By contrast, ARI and NMI stayed almost unchanged (ARI ~ 0.39 , NMI ~ 0.50). These supervised metrics stop improving once PCA retains around 90% of the variance, meaning that additional components mostly encode small variations that do not help label alignment.

2.2 K-Means with Reduced PCA Dimensions (Fixed $k = 10$)

To investigate whether stronger dimensionality reduction improves the intrinsic cluster structure, K-Means was re-evaluated on PCA dimensions from 10 to 100, with $k = 10$ fixed. From Figure 3, it is clear that at lower dimensions (10–30), K-Means achieves much stronger geometric separation, reflected in substantially higher Silhouette scores than in the 100–250 component range. As PCA dimensionality increases, Silhouette gradually declines, mirroring the trend observed in the high-dimensional experiments. However, ARI and NMI increase up to around 50 PCA components, after which improvements slow down, showing that moderate PCA preserves more digit-identity information without fully introducing the noise and overlap seen at very high dimensions.

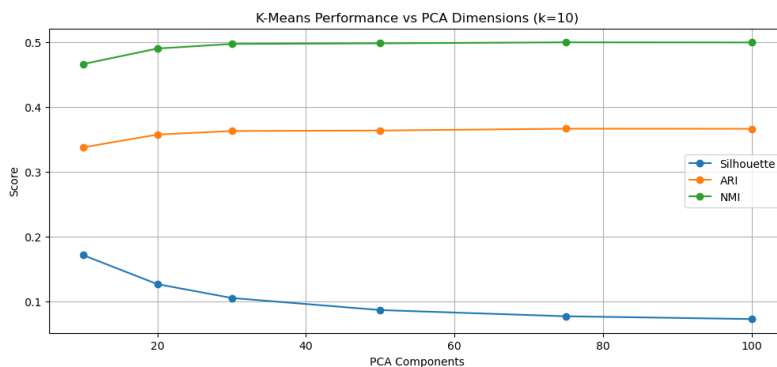


Figure 3: K-Means performance across reduced PCA dimensions.

PCA=20 was chosen because it achieves the highest silhouette and near-optimal ARI/NMI while keeping the model fast.

Parameter	Value
PCA components	20
algorithm	lloyd
init	k-means++
n_clusters	10
n_init	10
max_iter	300

Final K-Means configuration.

Metric	Score
Silhouette	0.1267
ARI	0.3573
NMI	0.4900

Final K-Means performance.

3 Gaussian Mixture Models

3.1 High-Dimensional PCA Experiments (100–200)

Gaussian Mixture Models (GMMs) were first evaluated on PCA dimensions between 100 and 200.

Parameter	Values Tested
Covariance	full, diag
Init. method	kmeans, random
Components	8, 10, 12

GMM hyperparameter grid

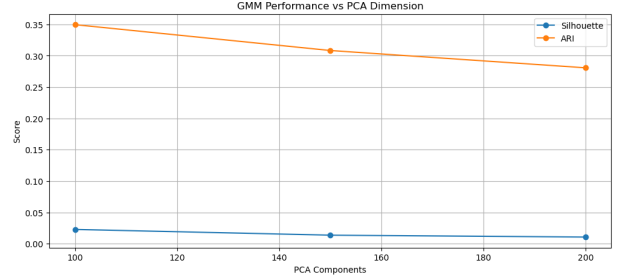


Figure 4: GMM performance across high PCA dimensions

Silhouette scores were extremely low for all settings (below 0.03), indicating heavy cluster overlap. This is expected because GMMs use *soft assignments*, which naturally lead to overlapping Gaussian densities. ARI and NMI degraded as PCA dimension increased, suggesting that higher PCA dimensions add variability that GMM struggles to model without overfitting.

3.2 Reduced PCA Dimensions (Fixed $n_components = 10$)

To test whether more aggressive compression benefits GMM, PCA dimensions from 10–100 were evaluated with a fixed 10-component mixture. Covariance types `full`, `diag`, and `tied` were used.

At very low dimensions (around 10), GMM achieves its strongest geometric separation, producing the highest Silhouette score observed across all settings. As dimensionality increases, Silhouette declines monotonically, reflecting the reintroduction of variance and overlap between Gaussian components. In contrast to the high-PCA regime where ARI and NMI deteriorated, here both metrics improve with increasing PCA dimension, peaking near 0.56 NMI at 100 components. This indicates that moderate-to-high PCA retains valuable digit-identity structure that GMM can model, even though the clusters become less geometrically distinct.

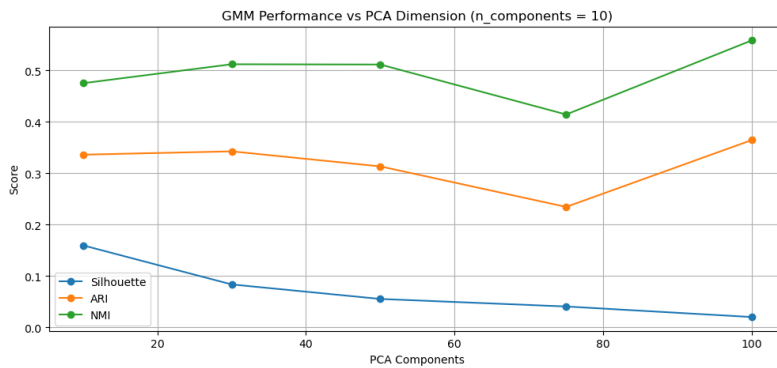


Figure 5: GMM performance across reduced PCA dimensions.

Although PCA=100 yields the highest NMI, it also significantly increases runtime and produces heavily overlapping clusters (very low Silhouette). PCA=20 was chosen as the final configuration because it provides a better balance between cluster compactness, label agreement, and computational efficiency.

Parameter	Value
PCA dimension	20
covariance_type	tied
init_params	kmeans
max_iter	200
n_components	10

Final GMM configuration.

Metric	Value
Silhouette	0.1077
ARI	0.3665
NMI	0.5039
Log-likelihood	-29.72

Final GMM performance.

4 Conclusion

4.1 Lessons from PCA Dimensionality

The experiments began with high PCA dimensions (100-250), but these proved unhelpful for both K-Means and GMM. Although high-dimensional PCA retains more variance, it also reintroduces noise-like fluctuations that weaken geometric separation and make Gaussian densities overlap heavily. This led to low Silhouette scores and unstable label alignment.

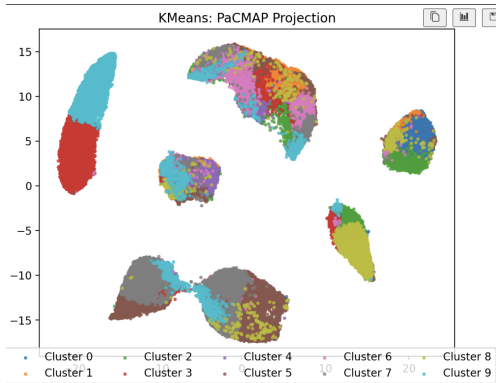
Reducing PCA revealed a clearer structure. Around 20 components, most pixel redundancy is removed while the essential digit shapes remain intact. This compression benefits both algorithms: clusters become more compact, PaCMAP visualisations stabilise, and ARI/NMI scores improve. Consequently, PCA = 20 was chosen as the final setting for both.

4.2 Choice of Cluster Number

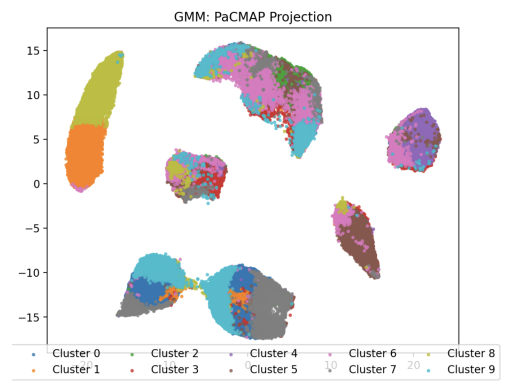
Initial sweeps suggested that K-Means preferred $k = 8$ while GMM performed slightly better with 12 components. However, MNIST is fundamentally a 10-class dataset. After retuning both methods at $k = 10$ and PCA = 20, the models became directly comparable and achieved more interpretable cluster assignments. This allowed a fairer evaluation of cluster separation, label consistency, and model behaviour.

4.3 Comparing Cluster Structure (PaCMAP and Samples)

PaCMAP projections for both models show remarkably similar global organisation once PCA is reduced to 20. K-Means forms slightly tighter, more clearly bounded groups, reflected in its higher Silhouette (0.1267). GMM shows softer boundaries because of probabilistic assignments, which explains its lower Silhouette (0.1077) but slightly higher ARI and NMI.



K-Means PaCMAP Projection

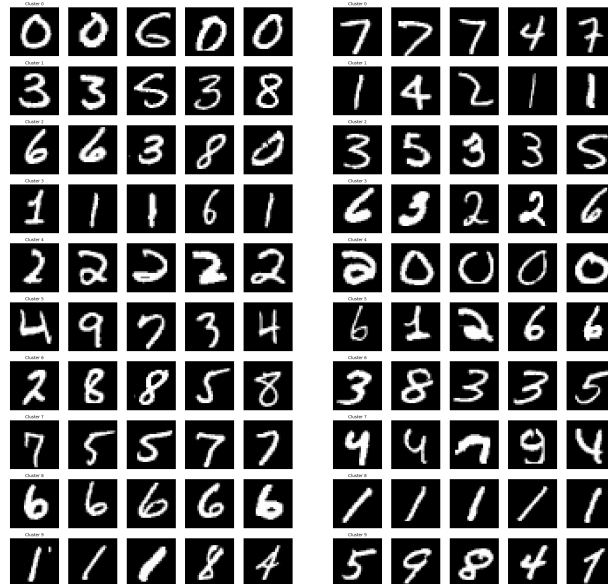


GMM PaCMAP Projection

4.4 Overall Assessment and Reflection

K-Means proved stronger in geometric compactness, runtime efficiency, and producing slightly sharper cluster prototypes, while GMM captured finer variations in digit shapes, giving marginally higher ARI and NMI. Interestingly, once PCA was reduced to 20 components, the two methods behaved far more similarly than expected: their PaCMAP projections aligned closely, their cluster centres looked alike, and their performance metrics differed only modestly. This process also highlighted a key lesson: higher dimensionality does not guarantee better clustering. Only after reducing PCA did MNIST's structure emerge cleanly. Both models separate easy digits well (0 and 1), while digits with strong shape variability (3, 5, 8) remain problematic. The sample images confirm this: both clusterers capture clean prototypes for 0, 1, and 2 but mix curved shapes such as 3/5/8 due to overlap in their stroke patterns.

Through iterative tuning (adjusting PCA, testing cluster counts, and validating with metrics and visualisations) both methods ultimately converged on stable, interpretable solutions.



K-Means sample cluster assignments (PCA=20)

GMM sample cluster assignments (PCA=20)